



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B. Sc. CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	INDUSTRIAL POLLUTION AND CONTROL				
Type of Course	VOCATIONAL MINOR				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites					
Course Summary	The course provides a comprehensive overview of basic concepts of industrial chemistry, Types of chemical industries, various industrial processes, intellectual property right and scope of industrial chemistry.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the adverse effects of emissions from chemical industries and guidelines set by the environmental protection agencies.	U	C	Seminar presentation /Assignment
CO2	Understand the causes and effects of air pollution.	Ap	P	Class test /Quiz /Assignment
CO3	To give an insight to various pollution control measures	An	P	Seminar Presentation / Instructor created exam
CO4	To familiarise waste water treatment methods	U	C	Instructor-created exams / Home Assignments
CO5	Explain different techniques for municipal solid waste and hazardous waste management	Ap	P	Assignment /Seminar presentation /Class test

CO6	Gain practical skills in some industrially important chemical analyses	Ap	P	Lab work /Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Industrial pollution		9	20
	1	Types of pollution	1	
	2	Types of emissions from chemical industries and effects of environment	2	
	3	Environment legislation	2	
	4	Effluent guidelines and standards	2	
	5	Sources and characteristics of pollutants in fertilizer, paper and pulp industry, petroleum and petroleum industry	2	
II	Air pollution		12	26
	6	Definition, Air quality, standards, emission standards, source and classification of air pollutants.	2	
	7	Major air pollutants – Oxides of carbon, nitrogen and sulphur	2	
	8	Particulates – London smog and photochemical smog	2	
	9	Air pollution control measures – Gravitational settling chamber, fabric filter	2	
	10	wet scrubber, catalytic converters, stacks and chimneys, cyclone collectors	2	
	11	Cottrell electrostatic precipitator, extraction ventilator, zoning and green belt.	2	
III	Water pollution		12	26
	12	Water quality parameters: DO, BOD and COD – Determination of BOD and COD	2	
	13	Toxic metals in water (Pb, Cd and Hg) – Minamata disaster (a brief study). Control of water pollution – Need for the protection of water bodies.	2	
	14	Sewage Treatment: Importance of sewage treatment, broad outline of sewage treatment (preliminary treatment, primary treatment, secondary or biological treatment disinfection	3	
	15	Sewage disposal methods of sewage, natural methods (dilution and land treatment)	3	

	16	miscellaneous treatments (oxidation ponds, aerated lagoons, oxidation ditch, anaerobic lagoons)	2	
IV	Soil pollution		12	26
	17	Soil pollution: Sources by industrial and urban wastes	2	
	18	Solid waste management - Sources and generation of solid wastes, their characterization, reduce-reuse-recycle paradigm	2	
	19	Chemical composition and classification of solid wastes	2	
	20	methods of disposal –sanitary landfill, secured land fill, incineration, pyrolysis, types of composting	2	
	21	Hospital waste and hazardous waste, rules regarding solid waste management	2	
	22	recycling of waste material, waste minimization technologies.	2	
V		1. Water analysis a. Determination of chemical oxygen demand (COD). b. Determination of biological oxygen demand (BOD) c. Estimation of Fluoride, Phosphate, Nitrate, Nitrite and Sulphate 2. Soil analysis a. Determination of TOC. b. Analysis of soil Sulphate. c. Determination of Ca²⁺ and Mg²⁺	30	

References:

1. Rao. C.S., “Environmental Pollution and Control Engineering”, 2nd Edition, Revised, New Age International, 2007
2. Mahajan. S.P., “Pollution Control in Process Industries”, Tata-McGraw Hill, New Delhi, 1985.
3. Narayana Rao, M. and Datta, A.K., “WasteWater Treatment”, 2nd Edition, Oxford and IBH Publications, New Delhi, 2005.
4. Swamy, A.V.N., “Industrial Pollution Control and Engineering”, Galgotia Publications, Hyderabad, 2005.
5. S. K. Banerjee, *Environmental Chemistry*, 2nd Edn., Prentice-Hall of India Pvt. Ltd., New Delhi, 2005.
6. A. K. De, *Environmental Chemistry*, 6th Edn., New Age International
7. N. P Cheremisinoff , *Handbook of Air Pollution Prevention and Control*, 2002.
8. M. Senapati, *Advanced Engineering Chemistry*, 2006.
9. K. C. Schiffner, *Air Pollution Control Equipment Selection Guide*, CRC Press, 2013.

10. K. B. Schnelle, C. A. Brown, *Air Pollution Control Technology Handbook*, CRC Press, 2016.

11. A. I. Vogel 'A Text Book of Quantitative Inorganic Analysis Including Elementary Instrumental Analysis': (Third Ed.) (ELBS)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	-	-	-	-	1	-	3	-	2	-	3	-	3
CO 2	-	-	-	-	1	-	3	-	2	-	3	-	3
CO 3	-	-	-	-	-	1	3	-	2	-	3	-	3
CO 4	-	-	-	-	-	1	3	-	2	-	3	-	3
CO 5	-	-	-	-	-	1	3	-	2	-	3	-	3
CO 6	-	-	2	-	-	2	3	-	2	-	3	-	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment / Discussion / Seminar
- Internal Examination
- Practical Examination
- Final Examination (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓	✓		✓
CO 5		✓	✓	
CO 6	✓			✓